WORKED SOLUTION

STEP I 2015, Question 4

Pure • Euclidean geometry

STEP I 2015, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q4

The key idea

Use the gradient of the curve as the parameter. If the tangent at the midpoint makes angle θ with the x-axis, the gradient there is $\tan\theta$, which fixes the midpoint. Each end is then a distance b from the midpoint along the unit vector $(\cos\theta, \sin\theta)$. For the area, the half-rod AP lies along the tangent at P, so the region is just the gap between a parabola and one of its tangents, and that integrand is a perfect square.

Solution

The locus of an end

Let the midpoint be P and let the rod make angle θ with the positive x-axis. Since $y = \frac{1}{4}x^2$ has $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}x$, and the rod is tangent to the curve at P,

$$\tfrac{1}{2}x_P = \tan\theta, \qquad \text{so} \qquad P = (2\tan\theta,\ \tan^2\theta).$$

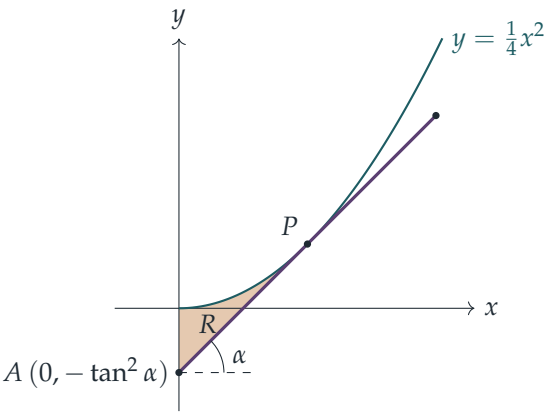
The condition $x \geq 0$ gives $\tan\theta \geq 0$, so we can take $0 \leq \theta < \frac{1}{2}\pi$. As θ runs over this range, $2\tan\theta$ takes every value in $[0, \infty)$, so every position of the midpoint arises from exactly one θ.

A unit vector along the rod is $(\cos\theta, \sin\theta)$, and each end is a distance b from P. So the ends are at $P \pm b(\cos\theta, \sin\theta)$. Taking the minus sign (the lower, left-hand end),

$$x = 2\tan\theta - b\cos\theta,$$
$$y = \tan^2\theta - b\sin\theta,$$

as required. (The other end traces out the same equations with $+b$ in place of $-b$.)

The area of R



The end that reaches the y-axis is the one with $x = 2\tan\theta - b\cos\theta$: the other end has $x = 2\tan\theta + b\cos\theta > 0$ always. So at $\theta = \alpha$,

$$2\tan\alpha = b\cos\alpha.$$

The y-coordinate of A is $\tan^2\alpha - b\sin\alpha$, and we can eliminate b using the last equation:

$$b\sin\alpha = (b\cos\alpha)\tan\alpha = 2\tan^2\alpha, \qquad \text{so} \qquad A = (0, -\tan^2\alpha).$$

Ask yourself Where is A relative to the origin, and what is the equation of the line AP?

A is below the origin, and the line AP is the tangent to the parabola at P. Write $t = \tan\alpha$. The tangent at $P = (2t, t^2)$ has gradient t, so its equation is

$$y = t^2 + t(x - 2t) = tx - t^2,$$

which indeed passes through $A = (0, -t^2)$. The parabola lies above this tangent, since

$$\tfrac{1}{4}x^2 - (tx - t^2) = (\tfrac{1}{2}x - t)^2 \geq 0.$$

So R is the set of points with $0 \leq x \leq 2t$ lying between the line AP (below) and the curve (above). Its left edge is the segment of the y-axis from A up to O. Therefore

$$\begin{aligned}\text{Area of } R &= \int_0^{2t} \left(\tfrac{1}{4}x^2 - (tx - t^2)\right) \mathrm{d}x = \int_0^{2t} (\tfrac{1}{2}x - t)^2 \,\mathrm{d}x \\ &= \left[\tfrac{2}{3}(\tfrac{1}{2}x - t)^3\right]_0^{2t} = 0 - \tfrac{2}{3}(-t)^3 = \tfrac{2}{3}t^3.\end{aligned}$$

So the area of R is $\frac{2}{3}\tan^3\alpha$, as required.

Notice that b has dropped out completely: the area depends only on where the tangent at P meets the y-axis. Of course b and α are linked by $2\sin\alpha = b\cos^2\alpha$, so b fixes α.

Checks

- Take $\alpha = \frac{1}{4}\pi$ (as in the diagram). Then $b\cos\alpha = 2$ gives $b = 2\sqrt{2}$, with $P = (2,1)$ and $A = (0,-1)$. The area is $\int_0^2 (\frac{1}{4}x^2 - x + 1)\,\mathrm{d}x = \frac{2}{3} - 2 + 2 = \frac{2}{3} = \frac{2}{3}\tan^3\frac{1}{4}\pi$.
- For $b = 2$, solving $2\tan\alpha = 2\cos\alpha$ numerically gives $\alpha \approx 0.6662$, and the area found by numerical integration is 0.32391, matching $\frac{2}{3}\tan^3\alpha$.
- As $b \to 0$, $\alpha \to 0$ and the region shrinks to nothing, as the formula predicts.

Common slips

- Splitting the region at the x-axis and forgetting the triangle below it. The single integral between the tangent and the curve handles both pieces at once.
- Using the wrong end of the rod: the end with $+b\cos\theta$ can never reach the y-axis.
- Leaving b in the answer instead of using $b\cos\alpha = 2\tan\alpha$ to eliminate it.

Beyond STEP

The midpoint traces the parabola, and each end traces a curve at constant distance b along the tangent. Building new curves from the tangents of an old one is a standard theme of the differential geometry of plane curves (involutes, envelopes and the tractrix are classic examples), met in first-year courses alongside arc length and curvature. The neat answer for the area recalls Archimedes' quadrature of the parabola, which shows that a parabolic segment has $\frac{4}{3}$ the area of a certain inscribed triangle. For more questions in this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.